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1.

Let $A = [a_{ij}]$, $B = [b_{ij}]$, $C = [c_{ij}]$ be $n \times n$ matrices, $c, d \in R$.

$A + B = [a_{ij} + b_{ij}]$ and $cA = [c \cdot a_{ij}]$ are both $n \times n$,

so the set is closed under addition and scalar multiplication.

1. $A + B = [a_{ij} + b_{ij}] = [b_{ij} + a_{ij}] = B + A$
2. $(A + B) + C = [(a_{ij} + b_{ij}) + c_{ij}] = [a_{ij} + (b_{ij} + c_{ij})] = A + (B + C)$
3. $O = [0]$: $A + O = [a_{ij} + 0] = A$
4. $-A = [-a_{ij}]$: $A + (-A) = [a_{ij} - a_{ij}] = O$
5. $c(A + B) = [c(a_{ij} + b_{ij})] = [c \cdot a_{ij} + c \cdot b_{ij}] = cA + cB$
6. $(c + d)A = [(c + d)a_{ij}] = [c \cdot a_{ij} + d \cdot a_{ij}] = cA + dA$
7. $c(dA) = [c(d \cdot a_{ij})] = [(cd)a_{ij}] = (cd)A$
8. $1A = [1 \cdot a_{ij}] = A$

All axioms hold, so the set of all $n \times n$ matrices is a vector space.

2.

$$\det \begin{Vmatrix} 1 & 0 & 4 & 5 \\ 3 & -2 & -1 & 5 \\ 4 & -3 & 1 & 6 \\ 0 & 5 & 3 & -1 \end{Vmatrix} = a_{11}M_{11} + a_{13}M_{13} - a_{14}M_{14}$$

$$M_{11} = -2[(1)(-1) - 6(3)] - (-1)[(-3)(-1) - 6(5)]$$
$$+ 5[(-3)(3) - (1)(5)] = -59$$

$$M_{13} = 3[(-3)(-1) - 6(5)] - (-2)[4(-1) - 6(0)]$$
$$+ 5[4(5) - (-3)(0)] = 11$$

$$M_{14} = 3[(-3)(3) - (1)(5)] - (-2)[4(3) - (1)(0)]$$

$+ (-1)[4(5) - (-3)(0)] = -38$

$\det = 1(-59) + 4(11) - 5(-38) = 175$

$\boxed{\det = 175 \neq 0}$

Linearly independent: $c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 = 0$ is a homogeneous system with $\det = 175 \neq 0$, so only $c_1 = c_2 = c_3 = c_4 = 0$.

Spanning: $\det \neq 0$, so $c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 = b$ has a solution for every $b \in R^4$.

S spans R^4 and is linearly independent, so S is a basis of R^4.

3. a.

$R_2 \rightarrow R_2 - 3R_1$

$= [3 - 3(1), 5 - 3(5), 0 - 3(-6), -2 - 3(6), -1 - 3(4)]$

$= [0, -10, 18, -20, -13]$

$$\begin{bmatrix} 1 & 5 & -6 & 6 & 4 \\ 0 & -10 & 18 & -20 & -13 \\ 0 & 0 & 0 & 1 & 5 \end{bmatrix}$$

Basis for the row space:

$\boxed{\{(1, 5, -6, 6, 4), (0, -10, 18, -20, -13), (0, 0, 0, 1, 5)\}}$

b.

Pivots are in columns 1, 2 and 4.

$$\boxed{\left\{ \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 \\ -2 \\ 1 \end{bmatrix} \right\}}$$

c.

$R_2 \to -(1/10)R_2 = [0, 1, -9/5, 2, 13/10]$

$R_1 \to R_1 - 5R_2$

$= [1 - 5(0), 5 - 5(1), -6 - 5(-9/5),$
$6 - 5(2), 4 - 5(13/10)]$

$= [1, 0, 3, -4, -5/2]$

$R_1 \to R_1 + 4R_3 = [1, 0, 3, 0, 35/2]$

$R_2 \to R_2 - 2R_3 = [0, 1, -9/5, 0, -87/10]$

$$\begin{pmatrix} 1 & 0 & 3 & 0 & 35/2 \\ 0 & 1 & -9/5 & 0 & -87/10 \\ 0 & 0 & 0 & 1 & 5 \end{pmatrix}$$

Let $x_3 = s$, $x_5 = t$.

$x_1 = -3s - (35/2)t$

$x_2 = (9/5)s + (87/10)t$

$x_4 = -5t$

$x = s(-3, 9/5, 1, 0, 0) + t(-35/2, 87/10, 0, -5, 1)$

Basis for the null space:

$\{(-15, 9, 5, 0, 0), (-175, 87, 0, -50, 10)\}$

d.

Rank = number of pivots = 3

nullity = 5 - 3 = 2

rank A = 3, nullity A = 2